EFFECTIVENESS OF DESCENT DATA WHOSE TOTAL SPACE IS A DISJOINT UNION OF SPECTRA OF ARTINIAN RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the equivalence relation and its finite classes are established in lemma-equivalence-classes-points and lemma-quasi-compact (cited at source lines 6950-6951); the open-and-closed preservation by phi is read off from the cartesian construction (source lines 6955-6959); the related lemma-equivalence-relation (defining the relation on points) was rejected as a card candidate for 'Details omitted.' - it is used here only as a cited input; no verification is omitted.

Editorial note. The label lemma-descent-disjoint-union-Artinian-along-fields names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Descent data as cartesian simplicial schemes. Same chapter, \label{lemma-cartesian-equivalent-descent-datum}. Proof lines 6941-6943: Y -> (X/S)_bullet is the cartesian morphism of simplicial schemes corresponding to (V, phi), with Y_0 = V.
- Equivalence classes of closed points with finite classes. Same chapter, \label{lemma-equivalence-classes-points} and \label{lemma-quasi-compact}. Proof lines 6946-6951: v_i the unique closed point of Spec(A_i); i ~ j iff v_i in T_{v_j}; by the two cited lemmas this is an equivalence relation with finite equivalence classes.
- Affineness of finite unions of Artinian spectra. Morphisms, Lemma \label{morphisms-lemma-Artinian-affine}. Proof lines 6960-6963: V_{i-bar} finite union of spectra of Artinian local rings, so V_{i-bar} -> X is affine.
- Effectiveness of affine descent data along fpqc coverings. Descent, Lemma \label{descent-lemma-affine}. Proof lines 6964-6971: {X -> S} is an fpqc covering, so each affine descent datum (V_{i-bar}, phi_{i-bar}) is effective.

Source reference. Original lemma 0245: sources/stacks-chow/tags.txt, source lines 2766–2766. Prerequisite pointer invoked at proof line(s) 6970; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 024C: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6472–6472. Prerequisite pointer invoked at proof line(s) 6944; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 024G: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6889–6889. Prerequisite pointer invoked at proof line(s) 6950; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 00E8: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6907–6907. Prerequisite pointer invoked at proof line(s) 6951; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01SI: sources/stacks-chow/tags.txt, source lines 2339–2339. Prerequisite pointer invoked at proof line(s) 6967; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $X \to S$ be a quasi-compact flat surjective morphism. Let (V, φ) be a descent datum relative to $X \to S$. If V is a disjoint union of spectra of Artinian rings, then (V, φ) is effective.*

Proof. Let $Y \to (X/S)_\bullet$ be the cartesian morphism of simplicial schemes corresponding to (V, φ) by Lemma 024C. Observe that $Y_0 = V$. Write $V = \coprod_{i \in I} \mathrm{Spec}(A_i)$ with each A_i local Artinian. Moreover, let $v_i \in V$ be the unique closed point of $\mathrm{Spec}(A_i)$ for all $i \in I$. Write $i \sim j$ if and only if $v_i \in T_{v_j}$ with notation as in Lemma 024G above. By Lemmas 024G and 00E8 this is an equivalence relation with finite equivalence classes. Let $\overline{I} = I/\sim$. Then we can write $V = \coprod_{\bar{i} \in \overline{I}} V_{\bar{i}}$ with $V_{\bar{i}} = \coprod_{i \in \bar{i}} \mathrm{Spec}(A_i)$. By construction we see that $\varphi : V \times_S X \to X \times_S V$ maps the open and closed subspaces $V_{\bar{i}} \times_S X$ into the open and closed subspaces $X \times_S V_{\bar{i}}$. In other words, we get descent data $(V_{\bar{i}}, \varphi_{\bar{i}})$, and (V, φ) is the coproduct of them in the category of descent data. Since each of the $V_{\bar{i}}$ is a finite union of spectra of Artinian local rings the morphism $V_{\bar{i}} \to X$ is affine, see Morphisms, Lemma 01SI. Since $\{X \to S\}$ is an fpqc covering we see that all the descent data $(V_{\bar{i}}, \varphi_{\bar{i}})$ are effective by Descent, Lemma 0245. $\square$